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AiM InfoTech

# YAMAHA
# MT-09 / FZ-09

## Release 1.00

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# 1 Models and years

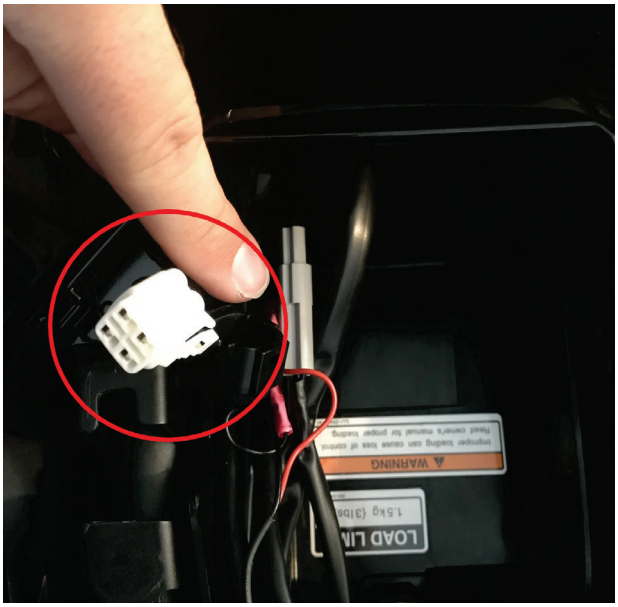
This document explains how to connect AiM devices to the vehicle Engine Control Unit (ECU) data stream.
Supported models and years are:

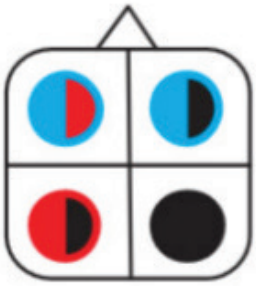
| | | |
|---|---|---|
| • Yamaha – FZ09 | MT09 | 2014 - 2020 |
| • Yamaha – FZ09 | MT09 | from 2021 (Euro5) |

**Warning**: for these models/years AiM recommends not to remove the stock dash, doing so will disable some of the bike's functions or safety controls. AiM Tech srl will not be held responsible for any consequences that may result from the replacement of the original instrumentation cluster.

# 2
# Wiring connection (2014 – 2020)

Yamaha MT-09 ECU features a bus communication protocol based on CAN that can be reached through the diagnostic connector (4pins Sumitomo connector), placed under the bike tail.
Below each diagnostic connector lead color and function is expressed.



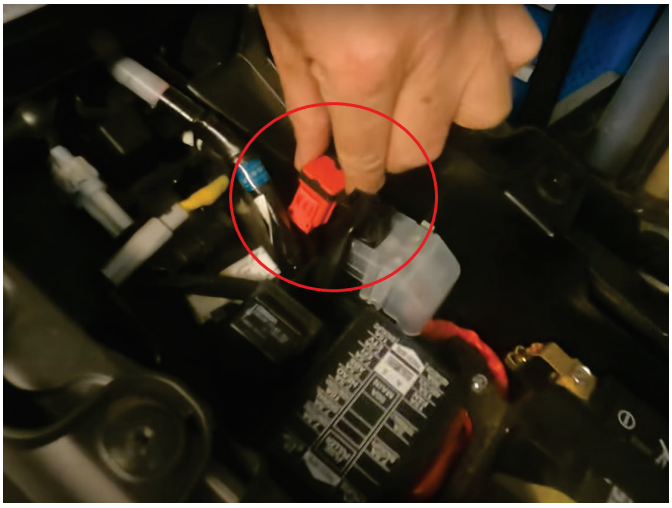


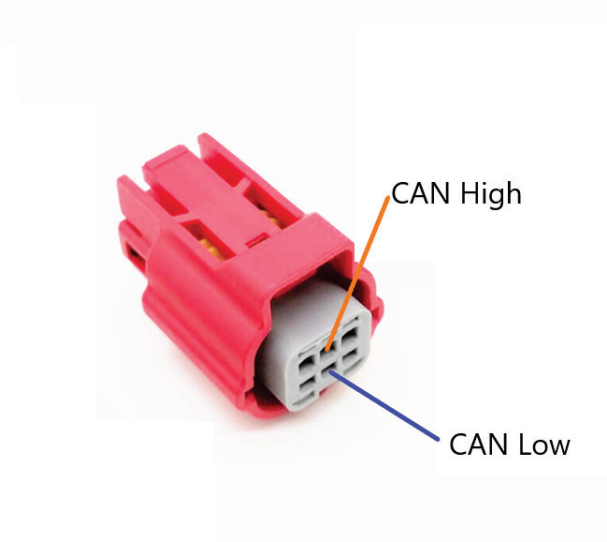
| Diagnostic connector | Function | AiM Cable | AiM color Cable |
|---|---|---|---|
| Blu/Red | CAN High | CAN + | White |
| Blu/Black | CAN Low | CAN - | Blu |
| Red/Black | +VB (unswitched) | V Batt | Red |
| Black | Ground | GND | Black |

| AiM supplies a dedicated cable with part number: | | |
|---|---|---|
| SOLO 2 DL | Cable length 1600mm | V02569340 |
| EVO4S | Cable length 500mm | V02585150 |

# 2.1
# Wiring connection (2021 onwards – Euro5)

Yamaha MT-09 Euro5 models are equipped with a bus communication protocol based on CAN that can be accessed through the standard OBDII Euro5 red connector placed under the bike seat.






| Diagnostic connector | Function | AiM Cable | AiM color Cable |
|---|---|---|---|
| 2 | CAN High | CAN+ | White |
| 3 | Ground | GND | Black |
| 4 | Voltage Battery | V Battery | Red |
| 5 | CAN Low | CAN- | Blue |

**AiM supplies a dedicated cable with part number:**

| | | |
|---|---|---|
| **Solo 2 DL** | **Cable length 1600 mm** | **“V02589130”** |
| **EVO4S** | **Cable length 400 mm** | **“V02585200”** |

# 3
# Race Studio configuration

Before connecting the AiM device to the ECU, set all functions using AiM software Race Studio. The parameters to set in the device configuration are:

- ECU manufacturer: **YAMAHA**
- ECU Model: **FZ09 MT09** (RS3 Only)

## 3.1
## OBDII - Race Studio configuration

Before connecting the AiM device to the ECU, set all functions using AiM software Race Studio. The parameters to set in the device configuration are:

- ECU manufacturer: **OBD_II**
- ECU model: **CAN** (Only EURO5 models)

# 4 Protocols

Channels received by AiM devices change according to the selected protocol.

## 4.1 ”YAMAHA – FZ09 MT09” protocol

Channels received by AiM devices configured with "YAMAHA – FZ07” protocol are:

| CHANNEL NAME | FUNCTION |
|---|---|
| RPM | Engine RPM |
| Speed | Vehicle speed |
| Speed | Vehicle speed 2 |
| Water Temp | Water temperature |
| Air Temp | Air temperature |
| PPS | Handle throttle position |
| TPS | Throttle position sensor |
| Gear | Engaged gear |
| Mode | Mode |
| FlagError | Flag error |
| 3= | MIL malfunction indicator lamp |
| Error | Error |

**Technical note**: not all data channels outlined in the ECU template are validated for each manufacturer’s model or variant; some of the outlined channels are model and year specific, and therefore may not be applicable.

# 4.2 ”OBDII - CAN” protocol

Channels received by AiM devices configured with "OBDII - CAN” protocol are:

| CHANNEL NAME | FUNCTION |
|---|---|
| OBDII RPM | Engine RPM |
| OBDII SPEED | Vehicle speed |
| OBDII ECT | Engine coolant temperature |
| OBDII TPS | Throttle position sensor |
| OBDII IAT | Intake air temperature |
| OBDII MAP | Manifold air pressure |
| OBDII MAF | Manifold air flow |
| OBDII FUEL LEVEL | Fuel level |
| OBDII PPS | Pedal position sensor |

**Technical note**: not all data channels outlined in the ECU template are validated for each manufacturer’s model or variant; some of the outlined channels are model and year specific, and therefore may not be applicable.